Introduction to Data Science

CSE 437: Data Science - Coding with Real World Data



What is Data Science?

Data science is a multidisciplinary blend of

- **data inference**
- **algorithm development**
- **technology**

In order to solve complex problems analytically.

Data science is to gain insights into the data through **computation**, **statistics**, and **visualization**.

Data Scientist

“A data scientist is someone who knows more statistics than a computer scientist and more computer science than a statistician.” - *Josh Blumenstock*

“Data scientist = statistician + programmer + coach + storyteller + artist” - *Shlomo Aragmon*

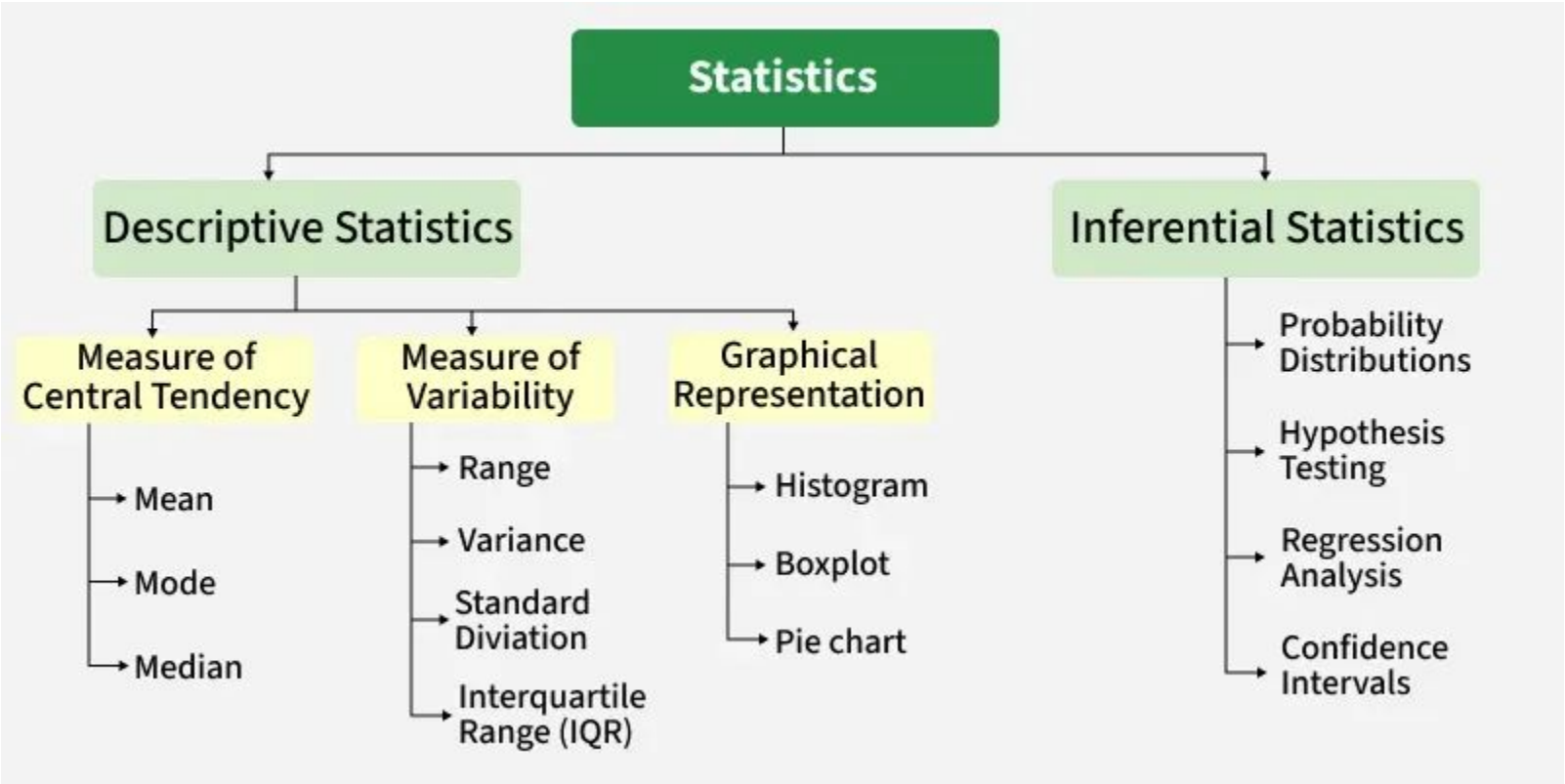
Phases of a Data Science Project

1. **Business problem:** asking questions, objectives, parameters
2. **Data acquisition:** databases, web servers, logs, APIs, online repositories
3. **Data preparation:**
    a. Data cleaning: inconsistent data/data types, misspelled attributes, missing values, duplicate values
    b. Transformation: mapping, quantization, binarization, normalization, scaling
4. **Exploratory data analysis:** dimensionality reduction, feature extraction, feature selection
5. **Data modeling:** machine learning to train and test to identify the best fit for business requirements (Python, R)
6. **Visualization and communication:** report generation with visualization, GUI
7. **Deployment and maintenance:** preproduction → production → real-time analytics → control and monitor performance

Big Data Sources

1. Online: click, ad, billing event etc
2. User-generated: FB, YouTube, X
3. Internet of things
4. Health or scientific computing

Statistics


Statistics
Descriptive Statistics
Inferential Statistics
Measure of Central Tendency
Measure of Variability
Graphical Representation
Mean
Mode
Median
Range
Variance
Standard Diviation
Interquartile Range (IQR)
Histogram
Boxplot
Pie chart
Probability Distributions
Hypothesis Testing
Regression Analysis
Confidence Intervals

Inferential Statistics

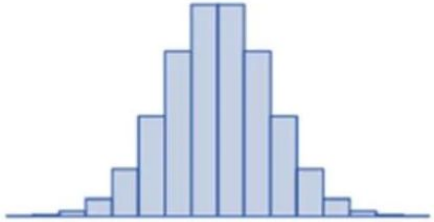
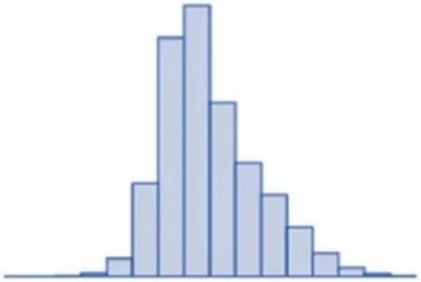
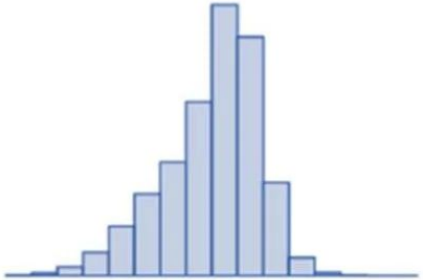
Inferential statistics allows us to make predictions and conclusions about a population based on sample data.

Need for inferential statistics:

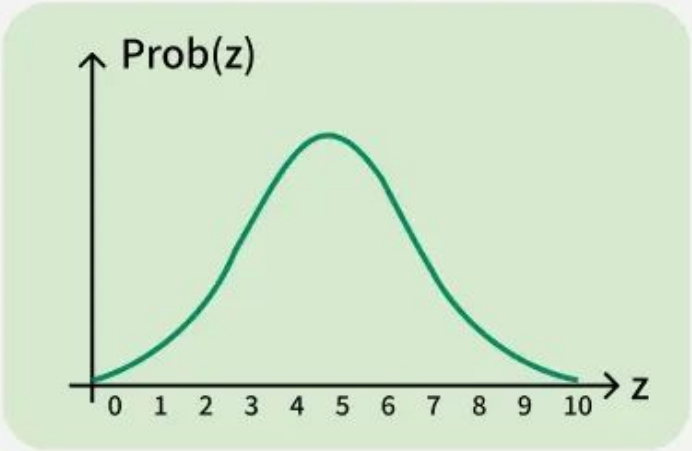
1. Analyzing an entire population is often impossible
2. Test claims or hypotheses
3. Calculate confidence intervals and p-values to measure uncertainty
4. Make predictions with statistical models

Data Distribution

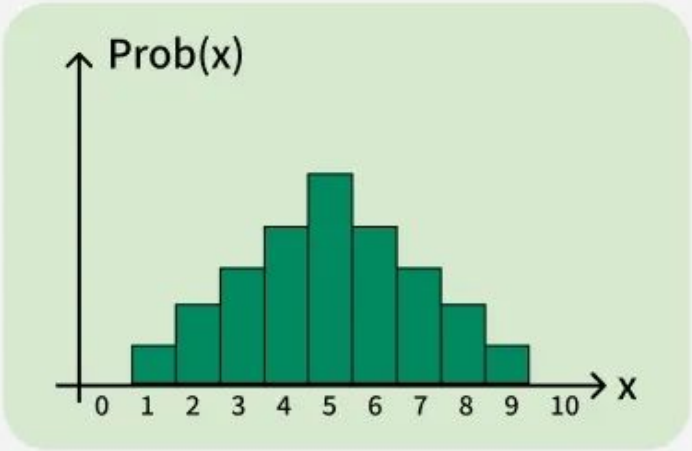
A statistical data distribution is a function that shows the possible values of a variable and how frequently they occur.

Data Distribution	Description
	Symmetric: A balanced distribution of both left and right tail where it's mean is approximately equal to the median.
	Skewed Right: An uneven frequency distribution where the mean value is larger than the median (median closer to the first quartile).
	Skewed Left: Opposite of the Skewed Right where median (median is closer to the third quartile) larger than mean.

Types of Statistical Data Distribution



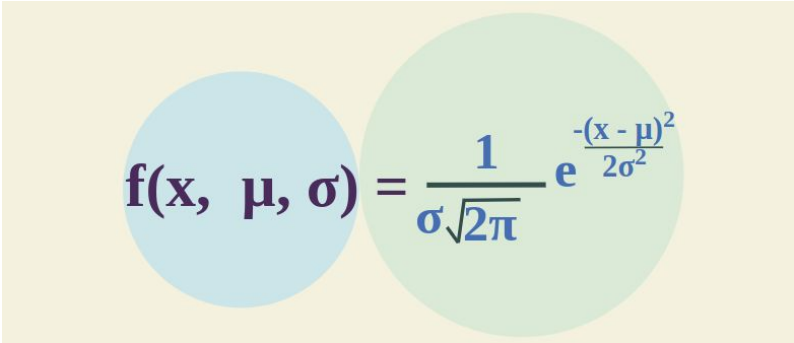
A **Continuous distribution** usually results from measuring something.



A **Discrete distribution** is based on random variables which can assume only clearly separated values.

Normal Distribution

- Also known as Gaussian distribution
- Mean = median = mode



Here, x is a random variable

μ is mean, σ is standard deviation

Normal Distribution Characteristics

- Symmetry - symmetric around its mean
- Mean = median = mode
- Bell-shaped curve
- Standard deviation - **empirical** rule (**68-95-99.7** rule) of data distribution

